



Resveratrol: potential therapeutic effects on ocular health

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Received: 7 July 2025 / Accepted: 1 September 2025 / Published online: 23 September 2025
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Abstract

Resveratrol, a polyphenolic compound predominantly found in grapes and red wine, exhibits significant biological activity due to its potent antioxidant and anti-inflammatory properties. Known for its cardioprotective, chemotherapeutic, neuroprotective, and anti-aging effects, resveratrol has garnered attention for its potential in managing age-related ocular diseases, such as glaucoma, cataract, diabetic retinopathy, and macular degeneration—conditions, where oxidative stress and inflammation play central roles in disease progression and vision impairment. Both in vitro and animal studies indicate that resveratrol impacts key biological pathways, including oxidative stress modulation, inflammation reduction, mitochondrial protection, apoptosis regulation, and angiogenesis suppression, which are implicated in the pathology of these ocular conditions. Resveratrol with antioxidant, anti-inflammatory, and anti-angiogenic properties, shows promise in treating ophthalmic diseases by targeting oxidative stress and abnormal angiogenesis, offering a safer, non-invasive alternative to conventional drugs and better scientific validation than herbal remedies. This review synthesizes current research on resveratrol's mechanisms of action, therapeutic potential, and limitations as a treatment for eye disorders, highlighting its promise as a nutritional intervention and areas requiring further exploration.

Keywords Age-related macular degeneration · Cataract · Diabetic retinopathy · Eye health · Glaucoma · Ocular disease · Phytochemicals

Introduction

Oxidative stress and inflammation are fundamental factors in the onset and development of age-related eye diseases, including glaucoma, cataracts, diabetic retinopathy, and macular degeneration, which, if left untreated, can result in progressive vision loss and blindness (Dib et al. 2015; Kowluru and Mishra 2015; Levkovitch-Verbin 2015). Emerging evidence suggests that phytochemicals, known for their antioxidant and anti-inflammatory properties, could play a role in preventing and managing these ocular conditions (Rhone and Basu 2008). Among these phytochemicals, resveratrol has garnered significant interest for its therapeutic potential in eye disease prevention.

Resveratrol (3,5,4'-trihydroxy-*trans*-stilbene) is a natural stilbenoid and a type of phenol known as a phytoalexin,

produced by several plants when they are injured or under pathogen attack, such as from bacteria or fungi (Langcake and Pryce 1977). It is found in foods, such as grape skins, blueberries, raspberries, and mulberries (Lippi et al. 2010). Interest in resveratrol rose with the concept of the “French Paradox,” which attributes the lower incidence of cardiovascular issues among the French, despite high dietary saturated fat, to the benefits of red wine, a rich source of non-flavonoid resveratrol (Richard et al. 2011; Wu and Hsieh 2011). Research indicates that resveratrol's antioxidant and anti-inflammatory properties might contribute to its cardioprotective, neuroprotective, chemotherapeutic, and anti-aging effects (De La Lastra and Villegas 2005; Gusman et al. 2001).

This review summarizes current scientific knowledge on resveratrol, examining its potential mechanisms of action, therapeutic applications, and limitations in treating eye disorders. A literature search using PubMed, Web of Knowledge, and Google was conducted for articles published up to 2015, using keywords, such as “resveratrol,” “eye disease,” “glaucoma,” “cataract,” and “anti-inflammatory.” Relevant data were included to focus on the objectives of this review.

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Sources

Food sources

Resveratrol is a polyphenolic compound in various plants. Rich sources include grapes, red wine, grape juice, peanuts, cocoa, and berries, such as blueberries, bilberries, and cranberries (Baur and Sinclair 2006). While red wine is a noted source, other polyphenols in wine are present at much higher levels than resveratrol (Guerrero et al. 2009). Resveratrol's structure consists of two aromatic rings linked by a methylene bridge, and it exists in two forms, *cis*- and *trans*-resveratrol, with the *trans* form being dominant in grapes and grape juice (Aggarwal et al. 2004; Rhone and Basu 2008). Estimated resveratrol content in some foods and beverages is presented in Table 1.

Supplements

Resveratrol supplements, primarily derived from *Polygonum cuspidatum* (Japanese knotweed) root extracts, as well as red wine and grape extracts, are widely available (Aggarwal et al. 2004). These products vary in concentration, ranging from < 1 to 500 mg per dose, though their safety and effectiveness in chronic disease prevention remain under review.

In vitro experiments

Most in vitro studies involved cultures of human retinal pigment epithelial cells (ARPE-19) in 4 out of 11 cases, with one study focusing on human retinal endothelial cells (hRECs). In addition, one study used non-retinal cells—specifically, peripheral blood mononuclear cells (PBMCs) from patients with proliferative DR studies were conducted on primary retinal cell cultures: two studies used rat Müller cells, two used bovine retinal capillary endothelial cells (BRECs), and one used rat retinal endothelial cells (RREC).

Only 3 out of the 11 studies provided details on cell identification, and none of the studies using the established ARPE-19 cell line provided information on measures to prevent misidentification.

Diabetic retinopathy (DR) was simulated by culturing cells in high glucose conditions (25–33 mM glucose in 7 studies; one of these studies also included H₂O₂ exposure), or by exposure to PDGF-BB (20 ng/mL in 1 study) or CoCl₂ (100–1000 µM in 1 study). One study did not apply any DR-mimicking treatment [20–24]. Resveratrol concentrations tested ranged from 1 µM to 10 mM, though only two studies from the same research group tested concentrations exceeding 1 mM. Table 2 depicts the observed in vitro studies on the impact of resveratrol on the cultured retinal cells.

In vivo experiments

The animal studies analyzed focused primarily on laboratory rodents, with research conducted on mice (C57Bl/6 in three studies and transgenic mice strains, such as Nestin-Cre and Tie2-Cre in one study) and rats (including four studies on Wistar rats, three on Sprague–Dawley rats, and one on Dark Agouti rats). Notably, only four of these eleven studies used pigmented rodent strains.

To induce diabetic retinopathy (DR), the majority of studies (10 out of 11) administered a single dose of streptozotocin, typically via intraperitoneal or intravitreal injection at doses between 50 and 60 mg/kg. An exception to this was one study that used an oxygen-induced retinopathy model, where neonatal mice were exposed to 75% oxygen from postnatal day 7 (P7) to day 12 (P12). With the exception of one study, all examined changes across the entire retina, with one study referencing “eye tissues” broadly.

Resveratrol administration varied across studies, with nine using oral delivery (doses ranging from 5 to 400 mg/kg/day), one study applying intraperitoneal injection (5 mg/kg/day), and another using intravitreal injection (5 µL of either 0.1 µg/mL or 1 µg/mL solution). The bias risk assessment for these in vivo studies is presented in, while a detailed summary of the findings on resveratrol's effects in animal models of DR is provided in Table 3.

Pharmacokinetics

Resveratrol's pharmacokinetics—how the body absorbs, distributes, metabolizes, and eliminates the compound—pose certain challenges:

- **Absorption and bioavailability:** Resveratrol is rapidly absorbed after oral administration, but its bioavailability is low due to extensive metabolism in the liver and intestines (Antonetti et al. 2012; Méndez-Del Villar et al. 2014). Studies show that only a small fraction of orally

Table 1 Concentration of resveratrol in different food sources

Food Source	Concentration of resveratrol
Red wine	0.1–14.3 µg/mL
White wines	<0.1–2.1 µg/mL
Cranberry raw juice	~0.2 µg/mL
Grapes	0.16–3.54 µg/g
Blueberries	Up to ~0.032 µg/g
Bilberries	Up to ~0.016 µg/g
Peanuts	0.02–1.92 µg/g
Boiled peanuts	5.1 µg/g
100% Natural peanut butters	0.65 µg/g (average)
<i>Polygonum cuspidatum</i> herb	524 µg/g

Table 2 In vitro studies on *resveratrol*'s impact on cultured retinal cells

Author, year (country)	Cell type and source	Insult	Resveratrol concentration (Duration)	Techniques used	Key findings
Chan Chi-Ming, 2013 (Taiwan, USA)	ARPE-19 human retinal pigment epithelial cells obtained from Food Industry Research and Development Institute (Taiwan)	PDGF-BB (20 ng/mL for 30 min at 37 °C)	1, 3, or 10 μ M	ECIS migration assay, MTT assay, dot binding, WB, scratch wound healing assay	Resveratrol reduced PDGF-BB-induced migration and signaling, potentially affecting PDGFR β , PI3K/Akt, and MAPK pathways. No influence on RPE cell adhesion to fibronectin
Chang Yo-Chen, 2017 (Taiwan)	ARPE-19 cells from ATCC	Hypoxia-mimetic treatment with CoCl ₂ (100–1000 μ M)	20 μ M	IP, WB, gelatin zymography, ELISA, RT-PCR	Resveratrol lowered hypoxia-induced HMGB1 secretion, reducing oxidative and hypoxic stress while affecting angiogenic and fibrotic responses
Chen Yuhua, 2019 (China)	Rat retinal endothelial cells (RREC)	High glucose (30 mM for 7 days)	10–500 μ M (24 h)	MTT assay, WB, RT-PCR	Resveratrol did not affect cell viability at concentrations $\leq$ 100 μ M under normal glucose. Showed inflammation reduction and increased PON1 expression; reduced active caspase-3 with high glucose
Kowluru Renu A., 2014 (USA)	Bovine retinal capillary endothelial cells (BRECs)	H ₂ O ₂ exposure (250 μ M for 1 h) and high glucose (20 mM for 4 days)	25 μ M	IP, RT-PCR, WB, enzyme activity, ROS assay	Resveratrol countered high glucose-related Sirt1 inhibition, reducing p65 acetylation, MMP-9 promoter activity, and MMP-9 activation
Li Jun, 2017 (China)	Bovine retinal capillary endothelial cells	High glucose (30 mM)	1, 5, 10, or 20 μ M (48 h)	Flow cytometry, RT-PCR, WB	Reduced high glucose-induced ROS levels via AMPK/Sirt1/PGC-1 α pathway activation, limiting apoptosis
Liu Shulin, 2016 (China)	Peripheral blood mononuclear cells from 19 diabetic retinopathy patients and 20 controls	No insult	10 μ M (72 h)	ELISA, WB, RT-PCR	In diabetic patients' PBMCs, resveratrol elevated SIRT1 levels while reducing IL-17 expression
Losso Jack, 2010 (USA)	ARPE-19 cells from ATCC	High glucose (33 mM)	1.25–10 μ M (9 days)	Crystal violet assay, ELISA, WB, scrape-loading/dye transfer assay	Resveratrol countered hyperglycemia-induced inflammation, restored GJIC, downregulated IL-6, IL-8, and Cx43, and suppressed TGF- β , PKC β , and COX-2

Table 2 (continued)

Author, year (country)	Cell type and source	Insult	Resveratrol concentration (Duration)	Techniques used	Key findings
Shen Hongjie, 2015 (China)	hRECs from Angio-Proteomie (USA)	High glucose (33 mM for 72 h)	Pterostilbene (1 mM, 72 h)	MTT assay, ELISA, enzyme activity, ROS assay	Decreased oxidation, inflammation, and retinal cell proliferation, delaying diabetic retinopathy progression
Subramani Murali, 2017 (India)	ARPE-19 from Karolinska Institute (Sweden)	Bevacizumab (0.25 mg/mL for 2 h)	100 μ M (48 h)	Trypan blue, MTT, FLICA, RT-PCR, BrdU, IHC, WB, scratch assay	Reduced VEGFR-2 levels and activation, lowered VEGF-A by 50%, decreased RPE cell proliferation, and stabilized the blood–retinal barrier
Zeng Kaihong, 2016 (China)	Rat Müller cells	High glucose (25 mM for at least 3 days)	10, 20, or 30 mM	Glutamate uptake, enzyme activity, IHC, RT-PCR	Prevented reduction of glutamate transporter (GLAST) expression and uptake under high glucose
Zeng Kaihong, 2017 (China)	Rat Müller cells	High glucose (25 mM for at least 3 days)	10, 20, or 30 mM	RT-PCR, enzyme activity, IHC	Reduced apoptosis in retinal Müller cells by regulating miR-29b, lowering Bax and SPI, and raising Bcl-2; miR-29b inhibitor negated resveratrol's anti-apoptotic effect

ingested resveratrol reaches systemic circulation in an active form (Szkudelski and Szkudelska 2011).

- **Metabolism and elimination:** After absorption, resveratrol undergoes rapid conjugation, resulting in metabolites, such as resveratrol sulfate and glucuronide, which limit its bioavailability (Liu et al. 2016; Zeng et al. 2017). These metabolites are mostly eliminated through urine. Due to this rapid metabolism, resveratrol's half-life is relatively short, often lasting just a few hours (Chen et al. 2019; Zeng et al. 2016).
- **Delivery to ocular tissues:** Achieving effective concentrations in ocular tissues is challenging due to resveratrol's poor solubility, stability, and rapid elimination (Chang et al. 2017; Soufi et al. 2012). Innovative delivery methods, such as encapsulation in nanoparticles, micelles, or liposomes, are being researched to enhance ocular bioavailability and target specific eye tissues (Alfaras et al. 2010; La Porte et al. 2010).

Pharmacodynamics of resveratrol in ophthalmic disease

Resveratrol's pharmacodynamic properties—its effects on the body and mechanisms of action—are promising for eye health, particularly in combating oxidative stress, inflammation, and angiogenesis in various ophthalmic conditions.

- **Antioxidant effects:** Resveratrol is a potent antioxidant, reducing oxidative stress by scavenging free radicals and upregulating antioxidant enzymes, such as superoxide dismutase (SOD), glutathione peroxidase, and catalase (Amri et al. 2012; Smoliga and Blanchard 2014; Summerlin et al. 2015). This action is particularly beneficial for diseases, such as age-related macular degeneration (AMD) and diabetic retinopathy (DR) (Kapetanovic et al. 2011), where oxidative damage plays a significant role in retinal cell degeneration (Bertelli and Das 2009; Nonomura et al. 1963).
- **Anti-inflammatory properties:** Chronic inflammation contributes to many eye diseases. Resveratrol inhibits key inflammatory pathways, such as nuclear factor-kappa B (NF- κ B) and cyclooxygenase-2 (COX-2), thereby reducing cytokine production (Jang et al. 1997; Kim et al. 1999). This anti-inflammatory effect has been observed in animal models of DR and uveitis, where resveratrol reduces retinal inflammation and vascular permeability (Lançon et al. 2004; Maier-Salamon et al. 2006).
- **Anti-angiogenic effects:** Pathological angiogenesis, or abnormal blood vessel formation, is a common factor in AMD and proliferative DR. Resveratrol inhibits vascular endothelial growth factor (VEGF) expression, a crucial factor in angiogenesis, by modulating transcription factors, such as hypoxia-inducible factor-1 α (HIF-1 α) and

Table 3 Effects of resveratrol in animal models of diabetic retinopathy

Author, year (country)	Animal type	Sample size (n)	Model of disease	Resveratrol dose and route	Duration	Techniques used	Tissues studied	Key findings
Al-Hussaini Heba, 2018 (Kuwait)	Male Dark Agouti rats (16 weeks)	5–10 per group	Streptozotocin (60 mg/kg, single, i.p.)	5 mg/kg, i.p., daily (6 days/week), starting 2 days post-STZ	14 or 30 days	RT-PCR, microarrays, WB	RPE	Enhanced gene expression affected by type 1 diabetes (normalization of Lpl, Rdh12, etc. after long-term dosing, but not short term)
Chen Yuhua, 2019 (China)	Male Sprague-Dawley rats (14 weeks)	4–5 per group	Streptozotocin (60 mg/kg, single, i.v.)	Intravitreal injection (0.1–1 µg/mL in one eye) or i.v. (5–50 µg/kg/day)	12 weeks	RT-PCR, WB, ELISA, enzyme assays	Whole retina	Reduced apoptosis, decreased inflammation, and enhanced antioxidant enzyme (PON1) activity
Kim Young Hee, 2012 (Korea)	Male C57BL/6 mice (8 weeks)	4 per group	Streptozotocin (55 mg/kg for 5 days, i.p.)	20 mg/kg orally once daily, starting 1 month post-STZ	8 weeks	Fluorescein angiography, IHC, WB	Whole retina	Prevented early diabetic retinal vascular damage and normalized VEGF expression
Kubota Shunsuke, 2011 (Japan)	C57BL/6 mice (6 weeks, sex unlisted)	7–8 per group	Streptozotocin (60 mg/kg for 5 days, i.p.)	50 mg/kg orally (via gastric tube), 7 weeks after STZ	7 days	WB, ELISA, perfusion labeling	Whole retina	Inhibited leukocyte adhesion, regulated AMPK and SIRT1, and reduced NF-κB signaling
Michan Shaday, 2014 (USA, Mexico, Australia)	Nestin-Cre, Tie2-Cre, C57BL/6J neonatal mice	6–20 per group	Oxygen-induced retinopathy (75% O ₂ exposure from P7 to P12)	400 mg/kg of micro-nized resveratrol (SRT501) orally from P5 to P17	13 days	IHC, HE staining, RT-PCR, WB	Whole retina	No significant protection against retinopathy; mild effect on vaso-obliteration
Soufi Farhad Ghadiri, 2012 (Iran)	Male Wistar rats (320–350 g)	48	Streptozotocin (50 mg/kg, single, i.p.)	5 mg/kg orally, daily for 4 months	4 months	TBARS assay, ELISA	Whole retina	Anti-hyperglycemic, anti-oxidative effects; reduced inflammation and retinal layer disorganization
Soufi Farhad Ghadiri, 2015 (Iran)	Male Wistar rats (12 weeks, 320–350 g)	6–12 per group	Streptozotocin (50 mg/kg, single, i.p.)	5 mg/kg orally, daily for 4 months	4 months	RT-PCR, ELISA	Whole retina	Lowered retinal apoptosis, reduced NF-κB activation, and anti-inflammatory mediator expression
Yar Seda Atiye, 2012 (Turkey)	Male Wistar rats (3 months, 250–300 g)	6 per group	Streptozotocin (55 mg/kg, single, i.p.)	10 mg/kg orally, daily for 4 week post-STZ	8 weeks	RT-PCR, biochemical assays	Eye tissues	Inhibited eNOS, no major effect on VEGF or MMP-9

Table 3 (continued)

Author, year (country)	Animal type	Sample size (n)	Model of disease	Resveratrol dose and route	Duration	Techniques used	Tissues studied	Key findings
Zeng Kaihong, 2016 (China)	Sprague–Dawley rats (14 weeks, sex unlisted)	68 per group	Streptozotocin (60 mg/kg, single, i.p.)	5 or 10 mg/kg orally, duration 1–7 months	1, 3, 5, or 7 months	ERG, RT-PCR, WT	Whole retina	Resveratrol improved retinal function, porter upregulation, mitigated diabetes-induced ERG changes
Zeng Kaihong, 2017 (China)	Male Sprague–Dawley rats (14 weeks, sex unlisted)	68 per group	Streptozotocin (60 mg/kg, single, i.p.)	5 or 10 mg/kg orally, starting 3 day post-STZ	1, 3, 5, or 7 months	TUNEL staining, IHC, RT-PCR, WB	Whole retina	Reduced retinal cell apoptosis, plasma glucose, and fructosamine in diabetic rats

NF- κ B (Delmas et al. 2011; Murias et al. 2008). Anti-VEGF effects of resveratrol can prevent abnormal blood vessel growth, which is essential for managing wet AMD and proliferative DR (Roy et al. 2009).

- **Neuroprotective action:** In addition to protecting retinal pigment epithelial (RPE) cells, resveratrol has neuroprotective effects on photoreceptors, which are crucial in preventing vision loss in AMD and other retinal diseases (Annabi et al. 2012; Xia et al. 2010). It modulates pathways, such as SIRT1 and ERK/MAPK, promoting cell survival and reducing apoptosis in retinal cells (Bhatt et al. 2011; Xu et al. 2009).

Mechanisms of action in ophthalmic diseases

Age-related macular degeneration (AMD)

- **Pathology and impact of resveratrol:** Age-related macular degeneration (AMD) involves the degeneration of the retinal pigment epithelium (RPE) and the formation of drusen, which are yellow deposits under the retina (Csiszar et al. 2008; Kim et al. 2011). Resveratrol exhibits protective effects in RPE cells, primarily by reducing oxidative stress and inflammation—two key contributors to AMD pathology (Yang et al. 2008). By mitigating these stressors, resveratrol shows potential for slowing down the degenerative process in AMD (Mudò et al. 2012).
- **Angiogenesis inhibition:** In cases of wet AMD, resveratrol plays a role in inhibiting angiogenesis, the process by which new blood vessels form (Della-Morte et al. 2009; Gertz et al. 2012). It achieves this by suppressing the production of vascular endothelial growth factor (VEGF), a protein critical to choroidal neovascularization, the abnormal formation of blood vessels under the retina that characterizes wet AMD (Howitz et al. 2003; Price et al. 2012). This suppression of VEGF indicates that resveratrol could serve as a beneficial adjunct therapy alongside current anti-VEGF treatments for managing AMD (Borra et al. 2005; Park et al. 2012).

Diabetic retinopathy (DR)

- **Resveratrol's role in DR management:** Diabetic Retinopathy (DR) arises from chronic hyperglycemia, which leads to inflammation and vascular damage in the retina (Borra et al. 2005). Resveratrol's antioxidant properties counteract oxidative stress, reducing VEGF activity and, in turn, decreasing pathological angiogenesis and vascular permeability in the retina (Park et al. 2012). These effects suggest that resveratrol could be a valuable thera-

peutic option for managing DR progression (Rahman and Islam 2011).

- **Molecular mechanisms:** Resveratrol's protective action against DR extends to its molecular mechanisms, where it modulates the activity of PKC β , a protein involved in cellular responses to glucose (Sinclair and Guarente 2014). In addition, resveratrol reduces pro-inflammatory markers, such as IL-6 and COX-2, which are elevated in DR and contribute to retinal damage (Wallerath et al. 2002). Through these mechanisms, resveratrol may help preserve retinal structures affected by DR-induced inflammation and stress.

Other ophthalmic conditions

- **Cataracts and uveitis:** Resveratrol has demonstrated protective effects against cataract formation by reducing oxidative stress markers, such as malondialdehyde, a byproduct of lipid peroxidation (Sinclair and Guarente 2014). In cases of uveitis, an inflammatory condition affecting the retina, resveratrol helps mitigate inflammation and suppresses NF- κ B activation, a pathway involved in immune response and inflammation (Wallerath et al. 2002). These properties make resveratrol a promising compound for managing both cataracts and uveitis.
- **Retinopathy of prematurity (ROP):** In retinopathy of prematurity (ROP), a condition that affects premature infants, resveratrol's anti-angiogenic effects show potential therapeutic benefits (Tomé-Carneiro et al. 2013). Animal studies have indicated that resveratrol can inhibit abnormal blood vessel growth in ROP, with nitric oxide signaling identified as a key pathway (Johnson et al. 2011). This suggests that resveratrol could play a role in preventing or treating ROP in clinical settings.

Therapeutic potential of resveratrol in ophthalmic diseases

Resveratrol's therapeutic potential in ophthalmic diseases lies in its ability to combat oxidative stress, inflammation, and pathological angiogenesis—all of which are key contributors to common retinal and degenerative eye diseases (Johnson et al. 2011; Walter 2001). These properties make resveratrol a promising candidate for treatments aimed at reducing progression or even potentially preventing some vision-related disorders.

1. Age-related macular degeneration (AMD)

Resveratrol has shown potential in slowing the progression of age-related macular degeneration (AMD), a leading cause of blindness in the elderly. Its powerful antioxidant properties help reduce oxidative stress

in retinal cells, which is a major factor in AMD (Borra et al. 2005; Sinclair and Guarente 2014). By downregulating pro-inflammatory factors, such as NF- κ B and HIF-1 α , and inhibiting vascular endothelial growth factor (VEGF), resveratrol may prevent the formation of abnormal blood vessels that contribute to the “wet” form of AMD (Walter 2001). In addition, its neuroprotective effects protect retinal pigment epithelium (RPE) cells and photoreceptors from degeneration, which is crucial for preserving vision (Piver et al. 2001).

2. Diabetic retinopathy (DR)

Diabetic retinopathy (DR) results from damage to the retinal blood vessels caused by high blood glucose levels in diabetes (Chow et al. 2010). Resveratrol's anti-inflammatory and anti-VEGF properties have shown promise in animal models for reducing inflammation, vascular leakage, and abnormal vessel growth in DR. By reducing oxidative stress and enhancing antioxidant enzyme activity, resveratrol also helps protect the blood–retinal barrier (Sim et al. 2014). In diabetic animal models, resveratrol has improved oxidative markers and reduced retinal vascular degeneration, suggesting it could be beneficial in managing DR progression (Weinreb et al. 2014).

3. Retinopathies and other inflammatory eye diseases

Resveratrol's anti-inflammatory effects can help manage other inflammatory eye conditions, such as uveitis and retinopathy of prematurity (ROP) (Weinreb et al. 2014). In uveitis, resveratrol has been shown to reduce levels of inflammatory markers, such as MCP-1 and ICAM-1, while inhibiting NF- κ B activation (Luna et al. 2009). In ROP, resveratrol's ability to reduce VEGF expression and prevent pathological angiogenesis may protect against retinal damage in premature infants. Its action in these conditions makes resveratrol a promising adjunct therapy for reducing inflammation and maintaining ocular health (Sim et al. 2014).

4. Cataracts and oxidative eye damage

Studies have shown that resveratrol may help delay cataract formation, particularly in cases where oxidative damage is a primary factor (Luna et al. 2009). By boosting antioxidant defenses, such as glutathione, and reducing lipid peroxidation, resveratrol helps prevent cellular damage in the lens. Animal studies have demonstrated that resveratrol can reduce cataract incidence in models of selenite-induced cataracts, suggesting a protective effect that could potentially benefit those at risk for cataracts (Pirhan et al. 2016).

5. Tumor inhibition in ocular melanoma

Resveratrol has shown inhibitory effects on the growth of ocular melanoma and retinoblastoma cells in vitro. By inducing mitochondrial dysfunction, promoting apoptosis, and causing cell cycle arrest in tumor cells, resvera-

trol may be a valuable adjunct for treating ocular tumors (Lindsey et al. 2015). Although more research is needed, these early findings highlight its potential in oncology applications within ophthalmology.

Clinical studies and human trials

Limited clinical studies on resveratrol in ocular diseases have reported beneficial effects, including improved retinal function and structural restoration in AMD patients (Razali et al. 2015). However, robust evidence from randomized controlled trials is lacking (Clark and Wordinger 2009). Some of the challenges such as dosing, bioavailability, and variable response across individuals highlight the need for further investigation (Liu et al. 2013).

Effectiveness of resveratrol with alternative medicine

Resveratrol, a polyphenolic compound found in grapes and berries, has emerged as a promising therapeutic agent for ophthalmic diseases, such as age-related macular degeneration (AMD), diabetic retinopathy (DR), glaucoma, and cataracts. Its antioxidant, anti-inflammatory, and anti-angiogenic properties make it a viable alternative to conventional treatments, such as anti-VEGF injections and corticosteroids (Dib et al. 2015; Levkovitch-Verbin 2015; Rhone and Basu 2008). For instance, studies have shown that resveratrol suppresses VEGF secretion, reducing abnormal blood vessel growth in AMD, and mitigates endoplasmic reticulum stress to protect retinal function in DR (Sinclair and Guarente 2014). In addition, it has demonstrated protective effects against oxidative stress in trabecular meshwork cells in glaucoma and lens damage in cataracts. Unlike conventional drugs, resveratrol is non-invasive, more affordable, and carries fewer side effects, though its poor bioavailability poses challenges that require advanced formulation techniques (Razali et al. 2015).

Resveratrol stands apart from conventional antioxidants by not only scavenging reactive oxygen species (ROS) but also activating intrinsic defense pathways, such as SIRT1, AMPK, and Nrf2. This enhances mitochondrial function, energy metabolism, and endogenous antioxidant enzyme activity (SOD, CAT, and GPx). Beyond this, it improves vascular health through eNOS stimulation, reduced mitochondrial oxidative stress, modulation of platelet nitric oxide, and direct vasodilation yielding vasorelaxant, anti-platelet, anti-inflammatory, anti-proliferative, and anti-calcifying benefits. Its therapeutic potential is further strengthened when combined with quercetin, vitamin D3, ferulic acid, or omega-3 fatty acids, which improve bioavailability and broaden efficacy.

Clinically, resveratrol has a wide safety margin and is generally well-tolerated, especially in stabilized forms like trans-resveratrol. Reported side effects are mild and mostly gastrointestinal, though doses above 2.5 g/day may cause discomfort, and very high levels have shown renal toxicity in animal studies. Its inhibition of platelet aggregation and cytochrome P450 enzymes raises potential drug interaction risks with anticoagulants, statins, calcium channel blockers, and antihistamines. Caution is also advised in hormone-sensitive cancers due to its estrogen-like activity. While safe for long-term use at moderate doses, supplementation should be individualized and supervised, particularly in patients on chronic medication or with underlying health conditions (Constantinescu and Mihis 2023; Rodella et al. 2023; Wang et al. 2016).

Compared to alternative medicines like herbal remedies, resveratrol has benefits from a robust scientific foundation. Preclinical studies highlight its targeted mechanisms, such as reducing oxidative damage and inflammation, which directly address the pathological processes of eye diseases. While herbal treatments often rely on general wellness effects with limited scientific validation, resveratrol stands out as a natural compound with well-characterized efficacy. Its potential as an adjunct or replacement therapy could improve accessibility and safety for patients. However, further clinical trials are essential to optimize its dosing, delivery methods, and long-term effectiveness, paving the way for its integration into mainstream ophthalmic care.

Formulations of resveratrol

Resveratrol ocular formulations are mainly in preclinical development, with nanotechnology approaches addressing issues of instability, poor solubility, and low bioavailability while aiming to enhance surface retention and retinal penetration. Nanoparticles such as lecithin/chitosan (six-fold higher exposure), PEGylated chitosan (better tolerance and deeper penetration), and PLGA (sustained release to retinal cells) show strong potential. Nanosuspensions like carboxymethylcellulose improve dissolution, stability, and reduce endothelial proliferation, while micelles (SNEDDS, pluronic/casein, mitochondria-targeted) enhance solubility, uptake, and antimicrobial or mitochondrial activity. Liposomes, including trimethylated chitosan-coated types, effectively reach the retina in animal models. Sustained-release systems include thermos-responsive hydrogels providing multi-day delivery and drug-loaded contact lenses with continuous antioxidant and anti-inflammatory effects. Non-topical options such as oral supplements remain limited by < 1% bioavailability, and intravitreal injections offer direct retinal delivery but are invasive (Gonzalez-Perez et al. 2024; Kaštelan et al. 2025; Saha 2021).

Resveratrol shows promise for eye health through anti-oxidant, anti-inflammatory, neuroprotective, and anti-angiogenic effects, but its impact depends on delivery. Orally, < 1% remains active due to rapid gut and liver metabolism, and high doses risk gastrointestinal upset. Local delivery bypasses first-pass metabolism, improving intraocular levels. Topical drops are limited by poor solubility and rapid clearance, yet nano-carriers (nanoparticles, micelles, liposomes) enhance stability, penetration, and retention. For posterior eye diseases, intravitreal nano-formulations achieve therapeutic retinal levels, while systemic parenteral routes increase circulation but are invasive and less targeted.

Therapeutically, topical resveratrol improves tear film stability, goblet cell function, and reduces oxidative stress in dry eye disease; protects retinal ganglion cells and lowers intraocular pressure in glaucoma; curbs VEGF-driven neovascularization, fibrosis, and oxidative damage in age-related macular degeneration; reduces vascular leakage and inflammation in diabetic retinopathy via PKC β modulation; and strengthens lens epithelial defenses in cataracts through antioxidant enzyme activation. While oral use offers systemic protection, direct ocular delivery, especially nano-technology-based is most effective for achieving therapeutic levels in specific eye diseases (Accomasso et al. 2024; Shetty et al. 2023; Varela-Fernández et al. 2020).

Limitations

Resveratrol faces several limitations that impact its therapeutic application in ophthalmology. One major challenge is its low bioavailability; resveratrol is rapidly metabolized in the body, which limits the concentration that reaches target tissues and thus reduces its therapeutic efficacy. In addition, resveratrol shows inconsistent effects across different cell types and conditions, necessitating careful consideration when determining its potential for various ophthalmic diseases. Finally, despite promising results in preclinical studies, there is a lack of large-scale, double-blind, placebo-controlled human trials to comprehensively confirm resveratrol's efficacy in treating eye diseases. Further research is essential to address these gaps and validate resveratrol's role in clinical ophthalmology.

Future perspectives

To overcome the limitations of resveratrol, advanced delivery systems are being explored, including liposomes, nanoparticles, and implantable devices, which could enhance its ocular delivery and prolong its retention in eye tissues. In addition, identifying active metabolites of resveratrol may allow researchers to create more potent analogs that

maximize therapeutic effects. Combining resveratrol with other anti-inflammatory or antioxidant agents also holds promise, potentially boosting efficacy while enabling lower doses, which may reduce adverse side effects. These strategies could significantly improve the clinical potential of resveratrol in treating eye diseases.

Conclusion

Resveratrol holds substantial promise for treating various ophthalmic diseases due to its anti-oxidative, anti-inflammatory, and anti-angiogenic properties. However, limitations such as poor bioavailability and lack of human studies must be addressed. Future research should focus on advanced delivery methods and comprehensive clinical trials to fully realize resveratrol's therapeutic potential in ocular health. In conclusion, resveratrol represents a versatile and potent candidate for the treatment of ophthalmic diseases, given its multifaceted action on key pathological pathways. However, realizing its full therapeutic potential requires overcoming significant challenges related to bioavailability and establishing robust clinical evidence of efficacy. Continued advancements in ocular drug delivery, combined with rigorous clinical trials, could pave the way for resveratrol to become an integral part of therapeutic strategies in managing complex eye diseases.

Author contributions Marina Andavar researched the literature and conceived the idea of the review, edited and critically revised the final manuscript. Bennett Kurien Biju designed the review and collected, analyzed the data and wrote the manuscript. All the authors contributed to the development of the article and approved the submitted version.

Funding No funding was received for this study.

Data availability This is a review article, so no data were collected.

Declarations

Conflict of interest The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. Peer reviewers on this manuscript have no relevant financial or other relationships to disclose.

Ethical approval and informed consent This review article is based on previously conducted studies and does not involve any human participants or animals, beyond their involvement in the previously published articles we cite. Therefore, no ethical approval and informed consent is required for this study.

Publication enhancement No publication enhancement materials were used.

Permissions No third-party contents were used in our review article.

Trial registration This is a review article; therefore, it does not require any trial registration.

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